MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, A.C., TYPE MDY

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated (see table I for size).
Second - A separator at manufacturer's option.
Third - Polyvinyl chloride insulation, 105°C, black (see table I for wall thickness). Standard identification code applied by method 1.
Fourth - Nineteen conductors, cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1, from the center outward. Either nonfibrous or a combination of nonfibrous and fibrous fillers shall be used to form a firm, well rounded assembly.
Fifth - Binder tape applied helically with overlap.
Sixth - Special thermoplastic jacket.
Seventh - Braided metal armor.
Eighth - Special thermoplastic jacket (gray), (see table I for wall thickness). Cable surface marking required.

FSC 6145

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TABLE I. Details.

Type and size	Conductor Size AWG	Insulation-thickness (minimum) (inch)	Inner cable jacket thickness (nominal) (inch)	Diameter over armor (nominal) (inch)	Cable jacket thickness (nominal) (inch)	Overall diameter		Conductor resistance per 1000 feet (maximum) (ohms)	Insulation resistance (megohms)		Cold bending mandrel (inches)
						minimum (inches)	maximum (inches)		Conductor to conductor and conductor to armor	Armor to water	
MDY-6	12(class B)	0.028	0.050	1.000	0.060	1.120	1.190	1.68	100	50	15
MDY-14	9(class B)	.040	.050	1.380	.060	1.500	1.570	.8407	100	50	21
MDY-23	7(class B)	.052	.085	1.710	.075	1.880	1.960	.5284	100	50	25
MDY-40	4(class C)	.052	.065	1.960	.075	2.150	2.240	.2590	90	45	29
MDY-60	2(class D)	.052	.065	2.250	.075	2.425	2.525	.1620	75	35	33

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - at 25°C ohms/1000 feet, maximum..	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2500
Conductor to armor.....	2500
Armor to water.....	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor and conductor to armor.....	(see table I)
Armor to water.....	(see table I)
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-915, table XXII, for limits of water leakage (with outer jacket removed).....	No failure
Group B:	
Cold bending, cable - (see table I for mandrel diameter).....	No damage
Drip - at 95 ± 1°C.....	Zero
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	100
Inner and outer jacket (cable)	
Tensile strength - lbf/in^2, minimum.....	2300
Elongation - percent, minimum.....	250

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Elongation - percent of unaged, minimum..	90
Inner and outer jacket (cable)	
Air oven (168 hours at 136°C)	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum..	50
Permanence of printing (jacket) - cycles, minimum.....	250
Permanence of printing (conductor) - cycles, minimum....	50
Cable filler removability.....	No failure
Armor - conformance to material construction and coverage.....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Cable aging and compatibility (95 ± 3°C).....	No failure
Flame propagation.....	No failure
Outer jacket durometer (type A), hardness, minimum.....	80

UNIT ORDERING LENGTHS: 1000 feet (nominal).

Revision letters are not used to denote changes due to the extensiveness of the changes.

NOTE: Not for Air Force Use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0885-11)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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